

## Unit 112 - Field Data Review

Mohawk Valley West Slope demonstration. Method **Tractor**, 80.5 ac, BAF 20 variable-radius cruise on a 300 ft grid, 40 plots, 428 trees tallied. Simulated data with planted recording errors; see docs/methods.md.

**Table 1. Stand summary from the BAF 20 cruise, per acre**

| STAND DENSITY        |                                    | STAND INFO          |                                                |
|----------------------|------------------------------------|---------------------|------------------------------------------------|
| Basal area           | 214 sq ft/ac (SE 2.5 %)            | Type                | Sierran mixed conifer                          |
| Trees per acre       | 92                                 | CWHR size / density | 4 / D (cover 87 %)                             |
| QMD                  | 20.6 in                            | SDI                 | 279 (48 % of max 580)                          |
| SAWTIMBER (>= 10 in) |                                    | BIOMASS (< 10 in)   |                                                |
| BA   QMD             | 210 sq ft/ac   21.5 in             | BA   QMD            | 4 sq ft/ac   8.6 in                            |
| TPA                  | 84                                 | TPA                 | 9                                              |
| Volume               | 33.2 MBF/ac (6608 cu ft)           | Volume              | 1.4 green tons/ac (64 cu ft)                   |
| TREATMENT TARGET     |                                    | CRUISE DESIGN       |                                                |
| Leave                | 35 % of max SDI = ~156 sq ft/ac BA | Sampling error      | 5.1 % at 95 % (t = 2.02, n = 40)               |
| Species (BA)         | WF 66, PP 63, IC 36, DF 30, SP 19  | Stratum standard    | 40 % - MEETS; 40 plots vs 20-plot practice: ok |

Cubic volume = total-stem CVTS by species (PNW-FIA California equations, MacLean and Berger 1976, CARB 2011 compendium) net of defect; Scribner board feet at 5.02 bf/cu ft (Keegan et al. 2010, table 2, California 2000-2006); green tons at PP 45, WF 47, DF 38, SP 49, IC 45 lb/cu ft of wood (Miles and Smith 2009, NRS-38 table 1A). SDI in the summation form; maximum is the basal-area-weighted FVS Western Sierra species maximum (table 3.5.1, rev. 2025-09-23). Sampling error at 95 % confidence with Student's t; stratum standard 40 % for tree-measurement sales (FSH 2409.12 ch. 40, 41.1). Sources and URLs: cruise\_data.xlsx, Standards and assumptions.

**Table 2. Stand table by 4-inch DBH class, per acre**

| DBH class, in | Trees      | TPA         | BA           | cu ft       | MBF         |
|---------------|------------|-------------|--------------|-------------|-------------|
| 4-7.9         | 2          | 3.4         | 1.0          | 15          | 0.1         |
| 8-11.9        | 19         | 16.0        | 9.5          | 206         | 1.0         |
| 12-15.9       | 47         | 21.9        | 23.5         | 523         | 2.6         |
| 16-19.9       | 62         | 17.6        | 31.0         | 787         | 3.9         |
| 20-23.9       | 61         | 11.7        | 30.5         | 865         | 4.3         |
| 24-27.9       | 69         | 9.5         | 34.5         | 1112        | 5.6         |
| 28-31.9       | 51         | 5.2         | 25.5         | 869         | 4.4         |
| 32-35.9       | 42         | 3.4         | 21.0         | 735         | 3.7         |
| 36-39.9       | 28         | 1.8         | 14.0         | 566         | 2.8         |
| 40-43.9       | 12         | 0.6         | 6.0          | 235         | 1.2         |
| 44-47.9       | 12         | 0.5         | 6.0          | 246         | 1.2         |
| 48-51.9       | 9          | 0.3         | 4.5          | 199         | 1.0         |
| 52-55.9       | 9          | 0.3         | 4.5          | 202         | 1.0         |
| 56-59.9       | 1          | 0.0         | 0.5          | 23          | 0.1         |
| 60-63.9       | 4          | 0.1         | 2.0          | 88          | 0.4         |
| <b>All</b>    | <b>428</b> | <b>92.5</b> | <b>214.0</b> | <b>6672</b> | <b>33.5</b> |

**Table 3. Stock table by species, per acre**

| Species    | Trees      | TPA         | BA           | QMD         | cu ft       | MBF         |
|------------|------------|-------------|--------------|-------------|-------------|-------------|
| WF         | 131        | 27.0        | 65.5         | 21.1        | 2068        | 10.4        |
| PP         | 126        | 27.2        | 63.0         | 20.6        | 2148        | 10.8        |
| IC         | 72         | 15.3        | 36.0         | 20.7        | 966         | 4.8         |
| DF         | 61         | 14.7        | 30.5         | 19.5        | 904         | 4.5         |
| SP         | 38         | 8.3         | 19.0         | 20.5        | 585         | 2.9         |
| <b>All</b> | <b>428</b> | <b>92.5</b> | <b>214.0</b> | <b>20.6</b> | <b>6672</b> | <b>33.5</b> |

Trees = trees tallied in the cruise; TPA, BA (sq ft/ac), cubic feet and Scribner MBF per acre by BAF expansion. Species: PP ponderosa pine, WF white fir, DF Douglas-fir, SP sugar pine, IC incense-cedar.

Unit 112: plots on the 300 ft grid (orange = QA flag)

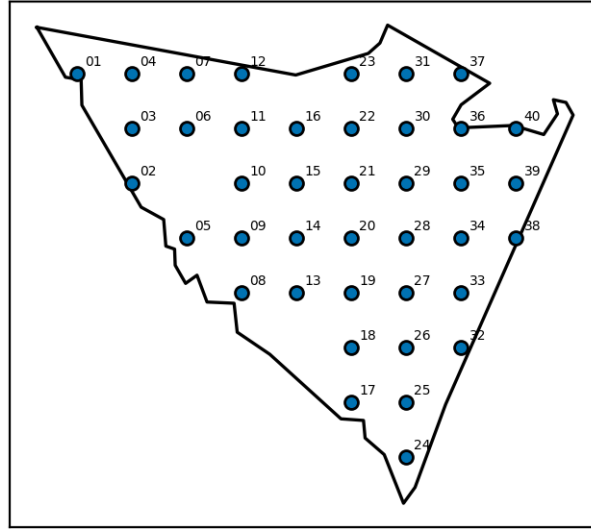


Figure 1. Plot locations with QA status.

**QA findings:** no issues found; unit accepted

Table 4. Plot list

| Plot   | Trees in | BA sq ft/ac | Slope % | Status |
|--------|----------|-------------|---------|--------|
| 112-01 | 11       | 220         | 12      | ok     |
| 112-02 | 11       | 220         | 10      | ok     |
| 112-03 | 11       | 220         | 17      | ok     |
| 112-04 | 12       | 240         | 12      | ok     |
| 112-05 | 9        | 180         | 21      | ok     |
| 112-06 | 12       | 240         | 36      | ok     |
| 112-07 | 10       | 200         | 10      | ok     |
| 112-08 | 5        | 100         | 21      | ok     |
| 112-09 | 12       | 240         | 10      | ok     |
| 112-10 | 11       | 220         | 10      | ok     |
| 112-11 | 9        | 180         | 10      | ok     |
| 112-12 | 11       | 220         | 29      | ok     |
| 112-13 | 10       | 200         | 12      | ok     |
| 112-14 | 13       | 260         | 8       | ok     |
| 112-15 | 13       | 260         | 15      | ok     |
| 112-16 | 13       | 260         | 16      | ok     |
| 112-17 | 10       | 200         | 13      | ok     |
| 112-18 | 12       | 240         | 36      | ok     |
| 112-19 | 9        | 180         | 13      | ok     |
| 112-20 | 11       | 220         | 21      | ok     |
| 112-21 | 10       | 200         | 14      | ok     |
| 112-22 | 13       | 260         | 13      | ok     |
| 112-23 | 10       | 200         | 22      | ok     |
| 112-24 | 13       | 260         | 28      | ok     |
| 112-25 | 10       | 200         | 26      | ok     |
| 112-26 | 12       | 240         | 11      | ok     |
| 112-27 | 8        | 160         | 9       | ok     |
| 112-28 | 10       | 200         | 15      | ok     |
| 112-29 | 8        | 160         | 13      | ok     |
| 112-30 | 11       | 220         | 12      | ok     |
| 112-31 | 8        | 160         | 7       | ok     |
| 112-32 | 12       | 240         | 25      | ok     |
| 112-33 | 11       | 220         | 26      | ok     |
| 112-34 | 11       | 220         | 20      | ok     |
| 112-35 | 10       | 200         | 10      | ok     |
| 112-36 | 13       | 260         | 39      | ok     |
| 112-37 | 11       | 220         | 16      | ok     |
| 112-38 | 9        | 180         | 16      | ok     |
| 112-39 | 12       | 240         | 19      | ok     |
| 112-40 | 11       | 220         | 30      | ok     |